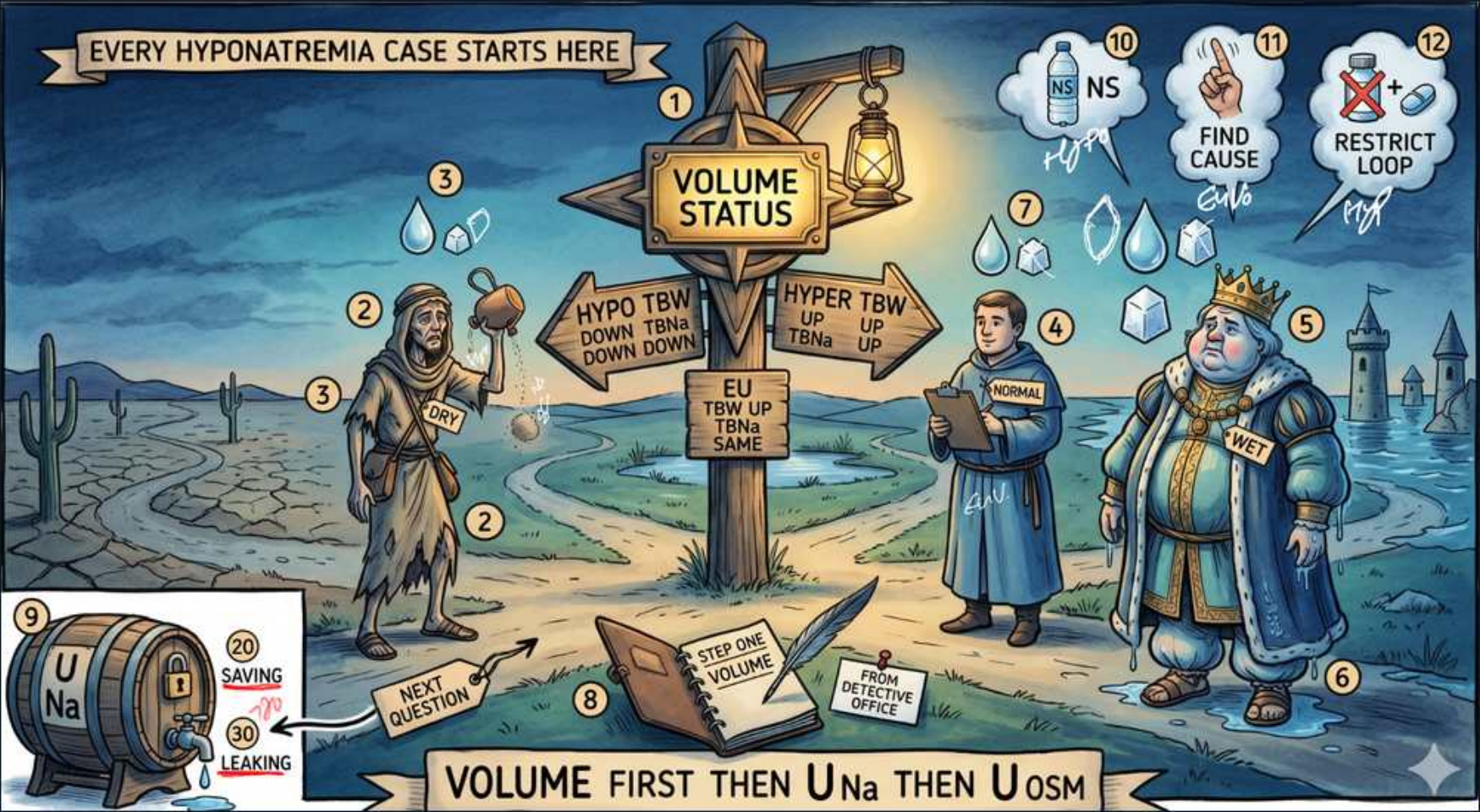


Hyponatremia — Volume Status Crossroads



1. Every case starts here

WHAT YOU SEE

Banner: EVERY HYPONATREMIA CASE STARTS HERE

WHAT IT ENCODES

Hyponatremia (Na <135) is a WATER problem, not a salt problem — nearly all true hyponatremia reflects excess free water relative to solute. Algorithm: (1) confirm true hypotonic hyponatremia by measuring serum osmolality (<275 mOsm/kg), excluding pseudohyponatremia (hyperlipidemia/paraproteins, normal osm) and hyperosmolar hyponatremia (hyperglycemia — correct Na up ~2.4 mEq per 100 mg/dL glucose >100; mannitol). (2) Then clinical volume assessment, which directs the entire differential. Mnemonic: 'Osm first, then volume, then urine.'

BOARD TRAP

WRONG: jumping to urine Na/osm before establishing serum tonicity and volume status. Discriminator — a hyperglycemic patient with Na 128 may have normal corrected Na (no true hyponatremia); measured serum osm separates true hypotonic hyponatremia from pseudo/hyperosmolar mimics. Volume status then frames how you read every urine value.

2. Volume status signpost

WHAT YOU SEE

Central wooden signpost labeled VOLUME STATUS

WHAT IT ENCODES

Volume status is the master fork: three branches — hypovolemic, euvolemic, hypervolemic — each defined by the pattern of total body water (TBW) vs total body sodium (TBNa). The unifying mechanism is ADH (vasopressin): low effective arterial blood volume (hypo- and hypervolemic states) drives non-osmotic ADH release, retaining free water and diluting Na. Euvolemic hyponatremia (SIADH) is ADH secreted inappropriately despite normal volume.

BOARD TRAP

WRONG: relying on volume exam alone — physical assessment of volume is notoriously insensitive (sensitivity ~50% for mild changes). Discriminator: pair the exam with urine Na, urine osm, and FENa rather than trusting JVD/edema in isolation, since early hypovolemia and euvolemia can look identical clinically.

3. Hypovolemic: down/down

WHAT YOU SEE

Sign: HYPO TBW DOWN TBNa DOWN DOWN

WHAT IT ENCODES

Hypovolemic hyponatremia: both TBW and total body Na fall, but Na falls MORE because losses (vomiting, diarrhea, diuretics, third-spacing) are partly replaced with hypotonic fluid/water. Low effective volume triggers ADH → water retention → dilutional drop in serum Na. Urine Na <20 (extrarenal loss, avid Na conservation) vs >20 (renal loss: thiazides, salt-wasting, mineralocorticoid deficiency). FENa <1% in extrarenal loss.

BOARD TRAP

WRONG: choosing 'thiazide started this week' but calling urine Na low — thiazide-induced hyponatremia causes RENAL Na wasting, so urine Na is typically >20–30 despite clinical dryness. Discriminator: thiazides (not loops) impair urinary dilution while preserving medullary concentrating ability, making them the classic diuretic cause of hyponatremia, especially in elderly women.

4. Euvolemic: TBW up, Na same

WHAT YOU SEE

Sign: EU TBW UP TBNa SAME

WHAT IT ENCODES

Euvolemic hyponatremia: free water retained (TBW up modestly) with near-normal total body Na — classic SIADH. Hallmark labs: urine osm >100 (often >300, inappropriately concentrated), urine Na >30, low serum uric acid (<4), low BUN, euvolemic exam. Causes: CNS disease, pulmonary disease (small-cell lung cancer = ectopic ADH), drugs (SSRIs, carbamazepine, cyclophosphamide), pain/nausea/post-op. ALWAYS exclude hypothyroidism and adrenal insufficiency, which mimic SIADH.

BOARD TRAP

WRONG: diagnosing SIADH without first ruling out hypothyroidism and cortisol deficiency — both raise ADH and produce identical euvolemic, concentrated-urine pictures. Discriminator: SIADH is a diagnosis of exclusion requiring normal TSH and a normal AM cortisol/cosyntropin response; missing Addison's risks giving fluid restriction when the patient actually needs hydrocortisone.

5. Hypervolemic: up/up

WHAT YOU SEE

Sign: HYPER TBW UP TBNa UP UP

WHAT IT ENCODES

Hypervolemic hyponatremia: both TBW and total body Na rise, but water rises MORE → dilution. Edematous states: heart failure, cirrhosis, nephrotic syndrome, advanced renal failure. Mechanism: low EFFECTIVE arterial volume (low cardiac output or splanchnic vasodilation) is sensed as hypovolemia → RAAS + ADH activation despite total-body overload. Urine Na <20 in HF/cirrhosis/nephrosis (avid renal Na retention) but >20 in renal failure (kidney can't conserve).

BOARD TRAP

WRONG: reading urine Na >20 in an edematous cirrhotic and concluding it isn't a sodium-avid state — discriminator: low urine Na (<20) is expected in HF/cirrhosis/nephrotic syndrome, but if the patient is on diuretics OR has intrinsic renal failure, urine Na rises and FENa loses specificity. Interpret urine Na before, not during, diuretic therapy.

6. Dry prophet (hypovolemic)

WHAT YOU SEE

Robed figure in cracked desert labeled DRY

WHAT IT ENCODES

Hypovolemic exam: dry mucous membranes, flat neck veins (low JVP), orthostatic hypotension/tachycardia, decreased skin turgor, oliguria. Low effective circulating volume → baroreceptor-mediated non-osmotic ADH release turns ADH ON, so the kidney retains water and urine is concentrated (Uosm high). Treatment restores volume to switch ADH off.

BOARD TRAP

WRONG: attributing concentrated urine here to SIADH — discriminator: in hypovolemia the concentrated urine is APPROPRIATE (volume defense), whereas in SIADH it is INAPPROPRIATE (euvolemic). The clinical clue is the dry exam + low urine Na (extrarenal); a saline challenge that corrects Na confirms hypovolemia and excludes SIADH.

7. Tear drops / losses

WHAT YOU SEE

Water-drop icons labeled near desert figure

WHAT IT ENCODES

Fluid-loss routes drive hypovolemic hyponatremia: GI (vomiting, diarrhea, NG suction), skin (sweating, burns), renal (diuretics, cerebral/renal salt wasting, osmotic diuresis), and third-spacing (pancreatitis, ileus). Urine Na is the splitter: <20 mEq/L = extrarenal loss with intact renal conservation; >20 mEq/L = renal loss. Replaced free water + ADH-driven retention produce the dilutional hyponatremia.

BOARD TRAP

WRONG: calling SIADH in a patient with subarachnoid hemorrhage and high urine Na — discriminator: cerebral salt wasting (CSW) is HYPOvolemic with renal Na loss and responds to salt + volume, whereas SIADH is euvoletic and worsens with saline. Volume status (CSW = dry, SIADH = euvoletic) is the key separator after neurosurgical/SAH events.

8. Normal-looking (euvoletic)

WHAT YOU SEE

Figure with clipboard labeled NORMAL

WHAT IT ENCODES

Euvoletic patients appear clinically normal — no edema, no overt dryness — so diagnosis hinges on labs: urine osm >100 (inappropriately concentrated), urine Na >30 , low uric acid, low BUN, plus exclusion of thyroid/adrenal disease and the mimics (primary polydipsia, low solute 'tea-and-toast'/beer potomania). In primary polydipsia and low-solute states, urine osm is LOW (<100) — the kidney is appropriately dilute.

BOARD TRAP

WRONG: assuming a normal exam plus concentrated urine equals SIADH — discriminator: urine osm <100 with hyponatremia points to primary polydipsia or beer potomania/low-solute intake (maximal dilution, water intake exceeds excretory capacity), NOT SIADH. Check urine osm to separate inappropriate concentration (SIADH) from appropriate dilution (water overload).

9. Wet king (hypervolemic)

WHAT YOU SEE

Crowned obese figure labeled WET

WHAT IT ENCODES

Hypervolemic exam: peripheral edema, ascites, elevated JVD, pulmonary crackles, S3, weight gain — total-body Na and water overload despite LOW serum Na. The paradox: total body Na is high yet effective arterial volume is low (poor forward flow), so ADH/RAAS stay activated and dilute the serum. Seen in HF, cirrhosis, nephrotic syndrome, renal failure.

BOARD TRAP

WRONG: giving normal saline to 'correct' the low Na in a decompensated HF patient — discriminator: these patients are already Na- and water-overloaded; saline worsens congestion. Treatment is water + salt restriction plus loop diuretics (and disease-specific therapy), not isotonic fluid. Reserve hypertonic saline strictly for severe symptomatic (seizing) hyponatremia.

10. NS for hypovolemia

WHAT YOU SEE

Thought bubble: NS bottle labeled NS

WHAT IT ENCODES

Hypovolemic hyponatremia treatment: isotonic 0.9% normal saline restores intravascular volume, which removes the hypovolemic stimulus for ADH → the kidney then dumps free water and Na rises. Correct slowly: max 6–8 mEq/L per 24h (some say ≤ 10) to avoid osmotic demyelination syndrome (ODS/central pontine myelinolysis). Watch for autocorrection/overshoot once ADH switches off.

BOARD TRAP

WRONG: giving NS to a SIADH/euvoletic patient — discriminator: in SIADH the urine is concentrated (Uosm often $>$ plasma osm $\sim 300+$), so the Na in the saline is excreted while the free water is RETAINED → serum Na paradoxically falls ('desalination'). When urine osm exceeds infused fluid osm, saline worsens SIADH; use fluid restriction $\pm$ salt tabs, loop, urea, or vaptans instead.

11. Find the cause (euvoletic)

WHAT YOU SEE

Thought bubble: FIND CAUSE with pointing hand

WHAT IT ENCODES

Euvoletic/SIADH management: hunt and treat the cause (stop offending drug — SSRIs, carbamazepine; treat pneumonia/CNS lesion; screen for small-cell lung cancer) PLUS fluid restriction (<800 – 1000 mL/day) as first-line. Adjuncts: salt tablets + loop diuretic, oral urea, or vasopressin-receptor antagonists (vaptans, e.g., tolvaptan) for refractory cases. Demeclocycline is largely historical.

BOARD TRAP

WRONG: starting tolvaptan or hypertonic saline as first-line for chronic asymptomatic SIADH — discriminator: fluid restriction is first-line; aggressive correction risks osmotic demyelination, and tolvaptan carries hepatotoxicity risk and can over-correct. Reserve rapid/hypertonic correction for acute severe symptomatic hyponatremia (seizures, coma): 100–150 mL 3% saline bolus targeting a 4–6 mEq/L rise.

12. Restrict / stop loop

WHAT YOU SEE

Thought bubble: RESTRICT LOOP with crossed pill

WHAT IT ENCODES

Hypervolemic hyponatremia treatment: fluid AND sodium restriction PLUS loop diuretics (furosemide) to excrete the excess Na and water; treat the underlying HF (guideline-directed therapy), cirrhosis, or nephrotic syndrome. Loops also blunt urinary concentrating ability, aiding free-water excretion. Vaptans/tolvaptan may help selected hyponatremic HF patients. Same slow-correction caution (≤ 6 –8 mEq/L/24h) applies.

BOARD TRAP

WRONG: using fluid restriction alone without diuresing the excess Na and water — discriminator: in hypervolemic states the problem is total-body Na+water overload, so restriction must be paired with loop diuresis to remove the load. Conversely, NEVER use saline here. Also avoid thiazides in this setting (they worsen hyponatremia by impairing dilution).

13. Step one: volume

WHAT YOU SEE

Open book labeled STEP ONE VOLUME, FROM DETECTIVE OFFICE

WHAT IT ENCODES

The detective sequence (mnemonic): VOLUME first → URINE Na → URINE OSM. Volume status narrows to the three branches; urine Na (<20 conserving vs >20 leaking) localizes renal vs extrarenal handling; urine osm (>100 inappropriate vs <100 appropriate dilution) separates ADH-driven retention from pure water overload. Add serum osm at the very start and uric acid/BUN to support SIADH.

BOARD TRAP

WRONG: ordering urine osm and FENa first and trying to back-calculate volume — discriminator: urine indices are only interpretable once volume status is set (e.g., urine Na <20 means avid conservation in BOTH hypovolemia and HF/cirrhosis). Anchor on volume first; otherwise identical urine numbers point to opposite diagnoses.

14. Urine Na barrel

WHAT YOU SEE

Barrel labeled U Na with locks, 20 SAVING / 30 LEAKING taps

WHAT IT ENCODES

Urine Na is the branch after volume: LOW (<20 mEq/L) = kidney avidly conserving Na (extrarenal/GI loss, or sodium-avid edematous states HF/cirrhosis/nephrotic); HIGH (>30 mEq/L) = renal Na loss/leak (diuretics, salt wasting, adrenal insufficiency) OR euvoletic SIADH (>30 with concentrated urine). Pair with FENa (<1% prerenal/avid; >1% renal/diuretic effect) — but FENa is unreliable on diuretics (use FEurea instead, <35% suggests avid state).

BOARD TRAP

WRONG: trusting a low FENa in a patient actively on furosemide — discriminator: loop and thiazide diuretics force Na into the urine and falsely RAISE FENa, masking an underlying prerenal/sodium-avid state. Use fractional excretion of urea (FEurea <35%) instead, since urea handling is less affected by diuretics, and interpret urine Na only off-diuretic when possible.